

K-Nearest Neighbor Algorithm

Dataset preparation:

Randomly split the dataset into Training (70%), Validation (15%), and

Test (15%)
`set Train_set=[], Val_set=[], Test_set=[]`

`//Following code shuffles your dataset list`

1. for each sample *S* in the dataset:
 2. generate a random number *R* in the range of $[0,1]$
 3. if $R \geq 0$ and $R \leq 0.7$
 4. append *S* in Train_set
 5. elif $R > 0.7$ and $R \leq 0.85$
 6. append *S* in Val_set
 7. else:
 8. append *S* in Test_set

KNN Classification:

Use Iris data [iris](#),

$K = 5$

1. for each sample *V* in the VALIDATION set:
 2. for each sample *T* in the TRAINING set:
 3. Find Euclidean distance between *V_x* (features- $\rightarrow$ N-1) and *T_x* (features- $\rightarrow$ N-1)
 4. Store *T* and the distance in list *L*
 5. Sort *L* in ascending order of distance
 6. Take the first *K* samples
 7. Take the majority class from the *K* samples (this is the detected class for sample *V*)
 8. Now, check if this class is correct or not

9. Calculate $\text{validation_accuracy} = (\text{correct VALIDATION samples}) / (\text{total VALIDATION samples}) * 100$

- Similarly, calculate validation accuracy for $K = 1, 3, 5, 10, 15$
- Make a table with 2 columns: K and Validation Accuracy ([report template](#))
- Now, take the K with the **highest** Validation Accuracy
- Use this best K to determine **Test Accuracy** (Simply replace the VALIDATION set with the TEST set)

KNN Regression:

Use diabetes data [diabetes](#)

Error = 0

1. for each sample V in the VALIDATION set:
 2. for each sample T in the TRAINING set:
 3. Find Euclidean distance between Vx and Tx
 4. Store Tx and the distance in list L
 5. Sort L in ascending order
 6. Take the first K samples
 7. Take the average output of the K samples (this is the determined output for sample V)
 8. Error = Error + (V true output - V determined output)²
 9. Calculate Mean_Squared_Error = Error/(total number of samples in VALIDATION set)
- Similarly, calculate Mean_Squared_Error for K = 1, 3, 5, 10, 15
 - Make a table with 2 columns: K and **Mean_Squared_Error** ([report template](#))
 - Now, take the K with the **minimum** Mean_Squared_Error
 - Use this best K to determine **Mean_Squared_Error for the Test set** (Simply replace the VALIDATION set with TEST set.

Instruction

- Submit the .ipynb file and a report ([report template](#)) .pdf file.
- **DO NOT USE LIBRARIES SUCH AS: "Sklearn", "Scikit learning" or "pandas" for this assignment.**